

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)MID SEMESTER EXAMINATION
DURATION: 1 HOURS 30 MINUTESWINTER SEMESTER, 2024-2025
FULL MARKS: 75**CSE 4503: Microprocessors and Assembly Language****Programmable calculators are not allowed. Do not write anything on the question paper.**Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Write an assembly language program structure to allocate exactly 64 Kbytes of memory for data segment and default memory bytes for stack segment. [Considering that the size for code segment may exceed 64 Kbytes.] 10
(CO4)
(PO1)
- b) Explain how a 20-bit address of a primary memory can be addressed by 16-bit register(s). Which register is the supporting registers of CS, DS, SS and ES registers in calculating the physical address? 4 + 6
(CO2)
(PO2)
- c) What is Implied Addressing? Explain the 20-bit address calculation process with appropriate examples for Direct and Indirect Addressing (i.e., both for 8-bit and 16-bit data addressing mode). 5
(CO2)
(PO2)
2. a) Suppose, Instruction Pointer (IP) of 8086 is moving in forward memory directions for executing all the following assembly language codes sequentially: 10
(CO1)
(PO1)


```
MOV AX, 1A2Bh;
NOT AX;
ADD AX, FFh;
```

After executing all the above arithmetic and logical instructions in single-step debug mode enabling interrupt option, what values would be there in AX and Flag (CF, PF, AF, ZF, SF, OF, IF, TF, and DF) registers?
- b) With appropriate examples, show the basic differences between the following instructions of 8086 microprocessor: 12
(CO4)
(PO1)
 - i. AND and TEST
 - ii. NOT and NEG
 - iii. SUB and CMP
- c) If DS = 0700H and BP = 0111H, then from the following instruction, derive the memory address(es) from which the data will be transferred: 3
(CO1)
(PO4)


```
MOV DX, [BP + F1E2H]
```
3. a) With appropriate example, differentiate between the operations of stack instructions and stack memory management for 8085 and 8086. Write your answer with respect to the use of SP register. 10
(CO2)
(PO2)
- b) Derive the machine codes of the following instructions using their instruction/coding template/format and also show how the machine codes of the instructions are to be stored in memory: 3 × 3
(CO1)
(PO2)
 - i. IN AL, ABH
 - ii. MOV DX, [BP + ABH]
 - iii. MOV CX, ABH

c) Derive the mistakes/errors of the following assembly language program and do the necessary corrections to re-write it:

(CO
(PO.

```
1  ORG 0100h
2  .DATA
3  P DB 10000001
4  Q DB 1A2BH
5  R DW 260
6  S DB ?
7
8  .CODE
9
10 MAIN PROC
11
12     MOV AL, P      AL = P
13     MOV BL, 2      BL = 2
14     MUL AL, BL
15     MOV CL, R
16     MOV S, Q
17
18     MAIN ENDP
19
20 END MAIN
21 RET
```